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Deep Learning and Applications

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Compressed Representation

- Sequential processing process data step by step.
 1. Learning temporal dependencies across the sequence.
 2. Each time step corresponds to one element in the input sequence.
 3. The model updates its hidden state to remember past information.
 4. Struggle to retain information across long sequences.

Capturing the global structure
and semantic essence.



- Compressed Representation
 1. Learn a compressed latent representation of the entire input.
 2. Capturing the **global structure** and **semantic essence**.





Compressed Representation

- **Supervised learning:** using labeled data,

$$\mathcal{D}_{\text{sup}} = \{(x^{(r)}, y^{(r)})\}_{r=1}^R$$

learn a mapping $f_{\theta}: x \rightarrow y$ that minimizes prediction error.

- **Unsupervised learning:** using unlabeled data,

$$\mathcal{D}_{\text{unsup}} = \{x^{(u)}\}_{u=1}^U$$

discover hidden patterns or representations in x without labels.

Reconstruction:

Transmitting or storing information efficiently.

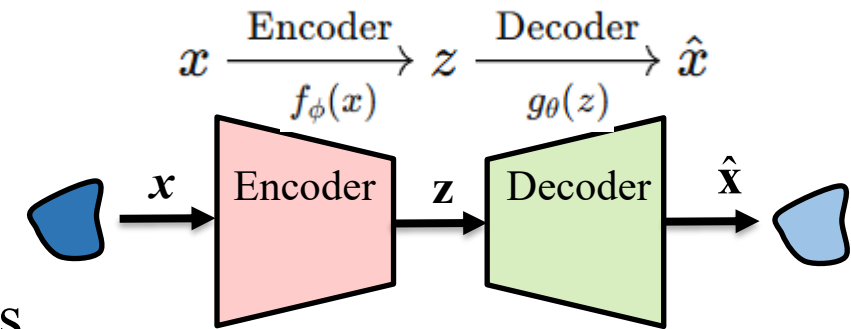


$$x \xrightarrow{\text{Encoder}} z \xrightarrow{\text{Decoder}} \hat{x}$$



Autoencoder (AE)

- Objective: make reconstruction $\hat{x}$ as close as possible to x .
- The Autoencoder **learns a compact encoding** of the data:
 1. **Encoder** extracts meaningful latent features z
 2. **Decoder** learns to **rebuild the full data** from those features.
- Loss function
 1. MSE: $L_{AE} = \|x - \hat{x}\|^2$ (Measures pixel-wise / value-wise difference)
 2. Cross-Entropy Loss:
$$L_{AE} = - \sum_i [x_i \log \hat{x}_i + (1 - x_i) \log(1 - \hat{x}_i)]$$
- Dimensionality Reduction (PCA, but nonlinear)
Learns to represent high-dimensional data with fewer variables.
- Feature Learning:
Automatically discovers useful patterns and features.





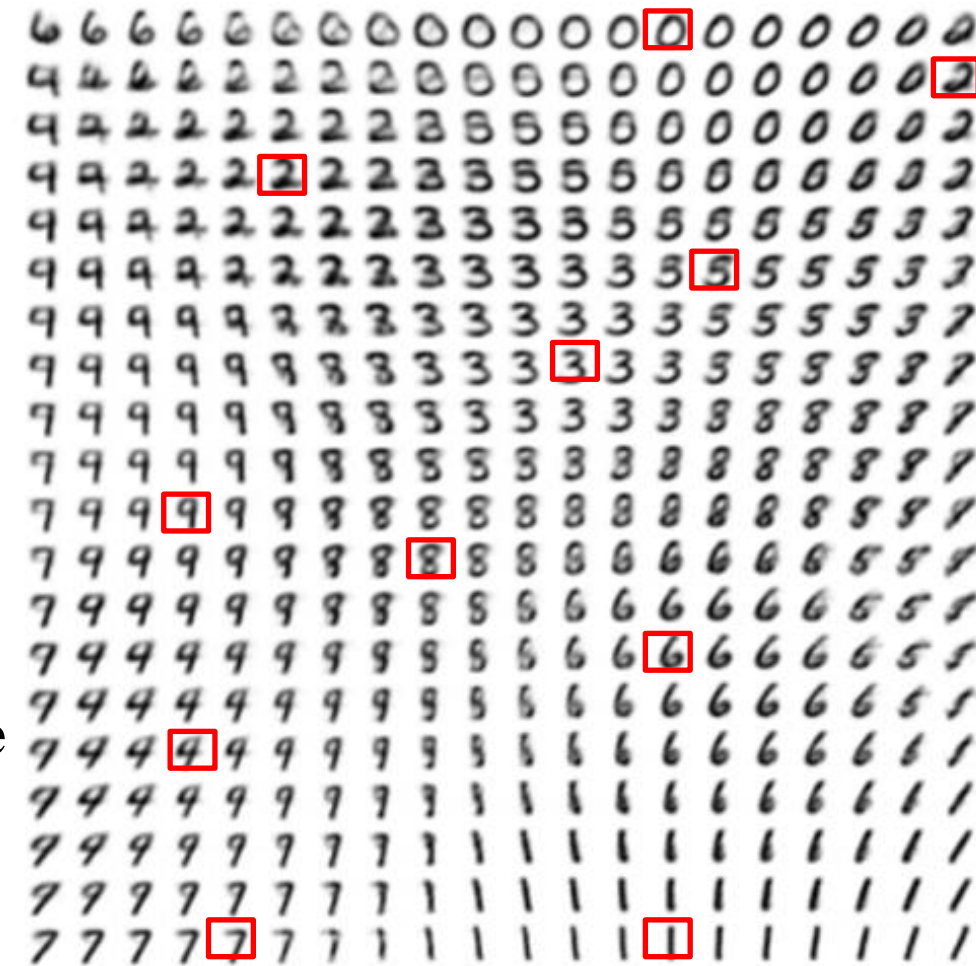
Autoencoder (AE)

- Limitations
 1. Unstructured Latent Space
 - a. The latent vector z has no meaningful organization or distribution.
 - b. The model does not learn a true latent probability distribution.
 2. Not Generative
 - a. Sampling a random z usually produces meaningless outputs.
 - b. The decoder cannot reliably generate new valid data samples.
 3. High Risk of Overfitting
 - a. The model may simply **memorize** the input data instead of learning the underlying data distribution.

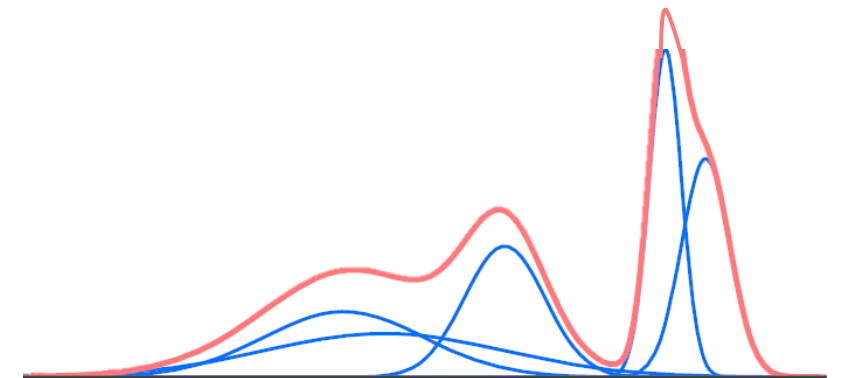


Latent Distribution

- Deterministic model
 1. AE only learns to **compress each input into a single vector**
— it does **not** learn the structure of the latent space.
 2. The latent space may be sparse, discontinuous, or fragmented
— only regions near training samples produce valid outputs.
 3. The latent space consists of **isolated points**
→ interpolation is possible mathematically, but **not guaranteed to be meaningful**.
- Generative model
 1. The latent space is constrained to follow a **continuous probability distribution**
 2. All samples are forced to **occupy a shared, smooth latent space**
→ dense, continuous, and topologically organized.
 3. Interpolation now corresponds to moving **along a valid data manifold**



(a) The two-dimensional map of the MNIST manifold





Variational Autoencoder (VAE)

- Goal: model the data distribution
 - The model should reproduce data that looks like the real thing.
 - Find model parameters θ so that the model assigns **high probability** to the kind of data we actually observe.

$$\max_{\theta} \log p_{\theta}(x)$$

maximum likelihood estimation (MLE)

Real data samples

$$x^{(1)}, x^{(2)}, \dots, x^{(N)} \sim p(x)$$

true but unknown data distribution

A generative model

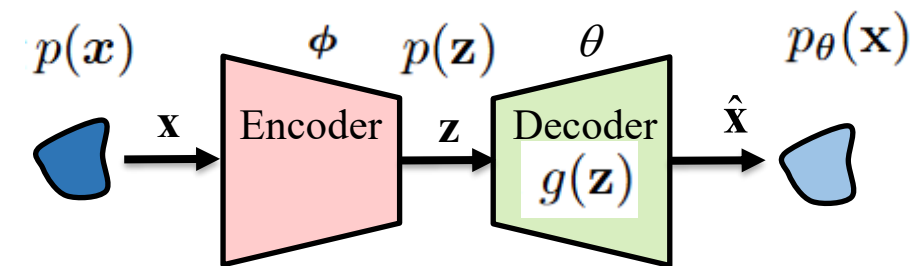
$$p_{\theta}(x) = \int p_{\theta}(x|z) p(z) dz = \int_{z \in \mathbb{R}^d} p_{\theta}(x | z) p(z) dz \quad \text{intractable}$$

how likely the model
thinks each possible
data point x is.

decoder or
likelihood
function

1. z in high dimensions

2. complicated neural
network output





Variational Autoencoder (VAE)

- Introduce a tractable, learnable approximation

$$q_{\phi}(z|x) \approx p_{\theta}(z|x)$$

encoder

- Goal of MLE: choose θ to make this large for data x .

1. Insert a harmless “1”

$$\log p_{\theta}(x) = \log p_{\theta}(x) \int q_{\phi}(z | x) dz.$$

2. Pull the constant inside the integral

$$= \int q_{\phi}(z | x) (\log p_{\theta}(x)) dz = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x)].$$

3. Replace p_{θ} with a ratio involving the joint and the posterior

$$= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{p_{\theta}(z | x)} \right] = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z) q_{\phi}(z | x)}{p_{\theta}(z | x) q_{\phi}(z | x)} \right]$$

$$\log p_{\theta}(x) = \underbrace{\mathbb{E}_{q_{\phi}} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z | x)} \right]}_{\mathcal{L}_{\text{ELBO}}(\theta, \phi)} + \underbrace{\mathbb{E}_{q_{\phi}} \left[\log \frac{q_{\phi}(z | x)}{p_{\theta}(z | x)} \right]}_{\geq 0}$$

$\mathcal{L}_{\text{ELBO}}(\theta, \phi) \leq \log p_{\theta}(x)$



Variational Autoencoder (VAE)

- **Kullback–Leibler divergence**, a measure of how different two probability distributions are.

$$D_{\text{KL}}(q_{\phi}(z|x) \parallel p_{\theta}(z|x)) = \mathbb{E}_{q_{\phi}}[\log q_{\phi}(z|x) - \log p_{\theta}(z|x)]$$

If they are identical $\rightarrow \text{KL} = 0$
If they differ $\rightarrow \text{KL} > 0$

The more q_{ϕ} diverges from the true posterior, the worse the bound becomes.

$$\log p_{\theta}(x) = \underbrace{\mathbb{E}_{q_{\phi}}[\log p_{\theta}(x, z) - \log q_{\phi}(z | x)]}_{\mathcal{L}_{\text{ELBO}}(\theta, \phi)} + \underbrace{D_{\text{KL}}(q_{\phi}(z | x) \parallel p_{\theta}(z | x))}_{\text{KL divergence} \geq 0}$$

optimize directly The remaining gap between our bound and the true log-likelihood

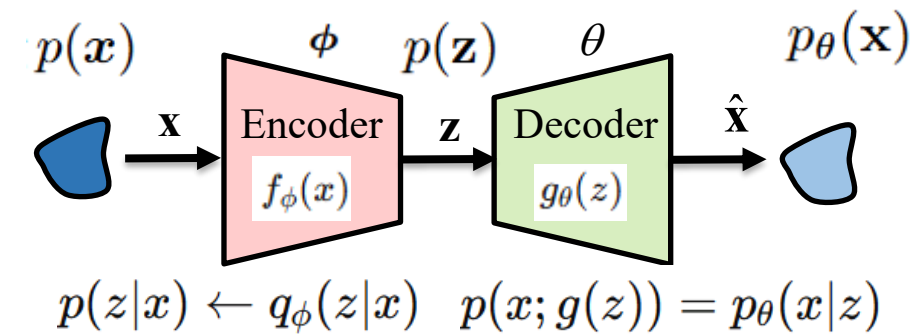
$\mathcal{L}_{\text{ELBO}} \leq \log p_{\theta}(x)$ literally a lower bound on the data likelihood.

$\rightarrow \max_{\theta, \phi} \mathcal{L}_{\text{ELBO}}(\theta, \phi)$ implicitly minimize that KL term.

$\rightarrow$ This ensures the encoder learns to approximate while the decoder learns to reconstruct data well.

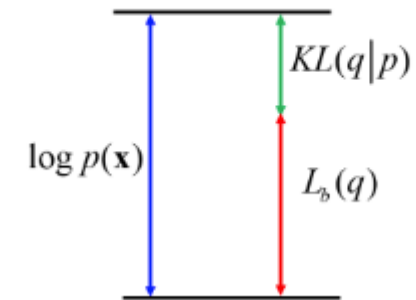
Variational Autoencoder (VAE)

- Evidence lower bound (ELBO)



$$\begin{aligned} \mathcal{L}_{\text{ELBO}}(\theta, \phi) &= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x, z)}{q_\phi(z|x)} \right] = \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x|z)p(z)}{q_\phi(z|x)} \right] \\ &= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] + \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(z)}{q_\phi(z|x)} \right] \\ &= \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_\phi(z|x) \| p(z))}_{\text{prior matching term}} \end{aligned}$$

Maximizes data likelihood
(Make output look like
input)

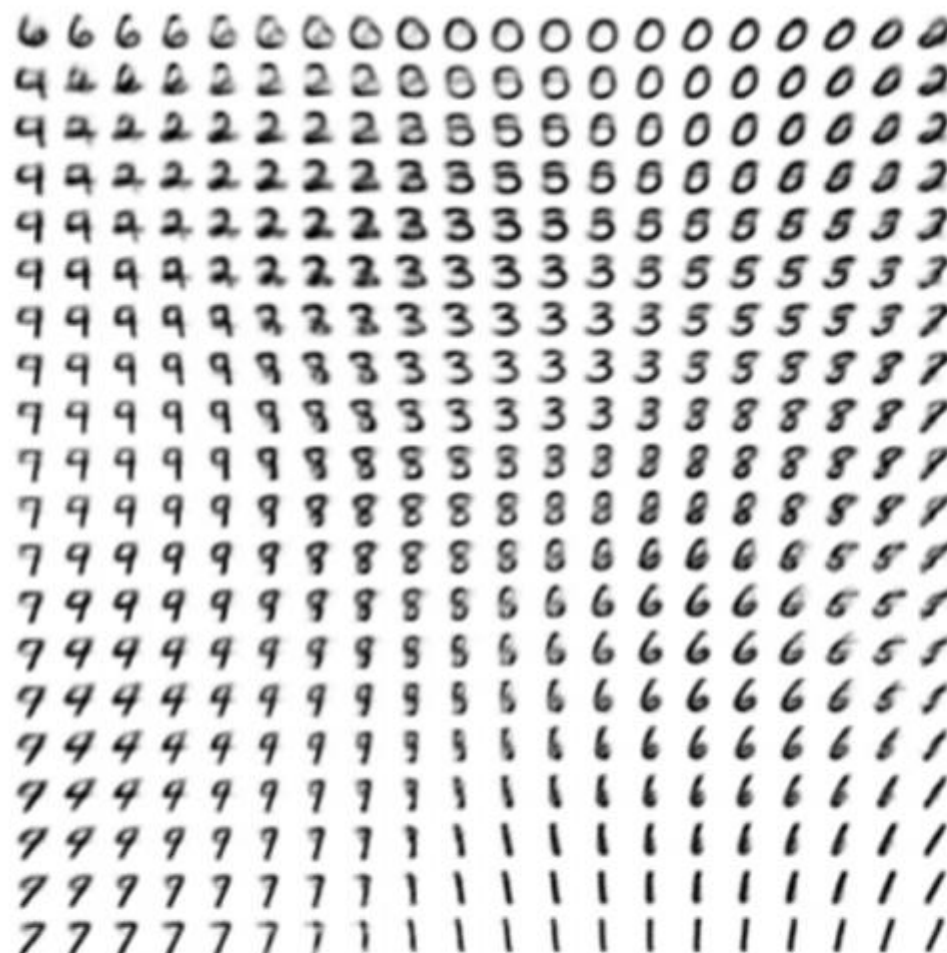


The distribution $q(z)$ is held fixed, and the lower bound L_b is maximized with respect to the parameter vector ϕ , yielding a revised value ϕ_{new} . Since the KL divergence is nonnegative, this process causes the log-likelihood $\log p(x|\phi)$ to increase by at least as much as the lower bound does.

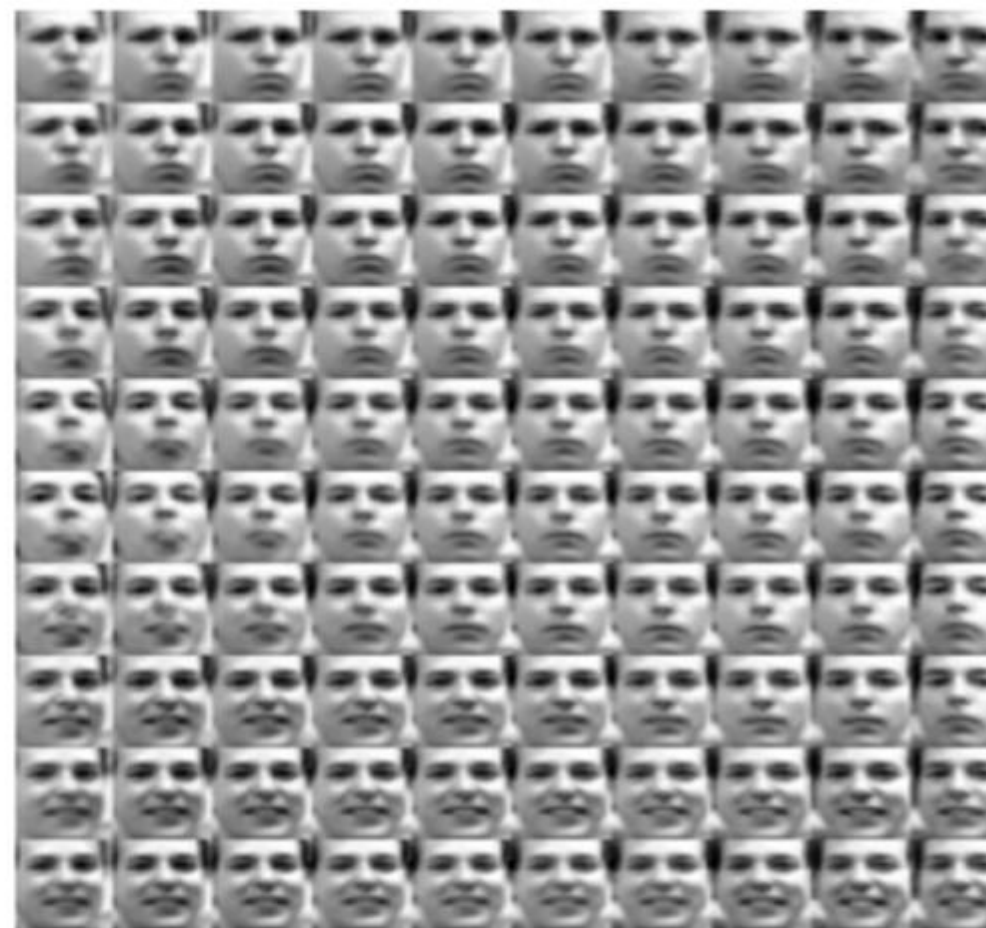
$$\begin{aligned} \log p_\theta(x) &= \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_\phi(z|x) \| p(z))}_{\text{prior matching term}} + \underbrace{D_{\text{KL}}(q_\phi(z|x) \| p_\theta(z|x))}_{\geq 0} \\ &\quad p(x; g(z)) = p_\theta(x|z) \quad p(z|x) \leftarrow q_\phi(z|x) \end{aligned}$$



Variational Autoencoder (VAE)



(a) The two-dimensional map of the MNIST manifold



(b) The two-dimensional map of the Frey faces manifold

Fig. 19: An understanding of how the model works by training a model with a 2-D latent code, even if we believe the intrinsic dimensionality of the data manifold is much higher. The images shown are not examples from the training set but images actually generated by the model $p(x|z)$, simply by changing the 2-D “code” z (each image corresponds to a different choice of “code” z on a 2-D uniform grid).